

<<Requirement>>

Autonomous System Vehicle

id=REQ=001

text= "The system shall autonomously navigate by utilizing line detection, obstacle avoidance, and color-based decision-making through onboard sensors. Navigation shall follow predefined routes with real-time adjustments based on environmental input."

<<Functional requirement>>
Navigation<<Functional requirement>>
Safety<<Requirement>>
id=002**Name: Speed Controller**

The Vehicle shall follow predefined routes (Oval or Eight)

<<Requirement>>
id=003**Name: Route Handling**

The vehicle adjusts its speed dynamically while turning or moving forward to ensure stable and efficient navigation.

<<Requirement>>
id=004**Name: Obstacle Proximity Stop Control**

text= "The system shall detect obstacles in the vehicle's path and initiate an immediate stop when the obstacle is within a 3 cm range."

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<<Requirement>>
id=005**Name: Performance Tuning**

text= "The speed must increase and decrease as needed."

<<Requirement>>
id=006**Name: Navigation in Oval**

text= "The car must navigate in an oval track."

<<Requirement>>
id=007**Name: Navigation in eight**

text= "The car must navigate in an eight."

<<Requirement>>
id=008**Name: Red Obstacle Detection**

text= "The Vehicle will Avoid the Obstacle."

<<Requirement>>
id=009**Name: Green Obstacle Detection**

text= "The Vehicle pushes the Obstacle."

<<Requirement>>
id=010**Name: Unknown Obstacle Parking Response**

text= "If the vehicle detects an obstacle of unknown or unrecognized color, it shall safety come to a stop and initiate a parking maneuver."